

	Case Name: Railway Bridge (3)	Sector	Construction (civil)	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
		Risk Analysis	Random simulation	
Submitted by	Mieke De Saedeleer		One of nine std. scenarios	
Date	2016		User defined distributions	
File Name	C2016-03 Railway Bridge (3)	Project Control	Automatic tracking	
			Tracking based on user input	

1. Project description

Project authenticity

This project involves the **replacement or rehabilitation of a railway bridge**. It begins with preparatory studies and site setup, followed by the **demolition of the existing structure**. The core phase focuses on **constructing the new bridge components**, including foundations and superstructure. Finally, the project concludes with finishing works and making the bridge ready for safe railway operations.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	27	Serial/Parallel (SP)	16%
Planned Duration (PD)	203	Activity Distribution (AD)	37%
Budget At Completion (BAC)	926888.01	Length of Arcs (LA)	0%
Renewable Resources	-	Topological Float (TF)	56%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	15%	15%	1.03
CRI-rho	19%	17%	0.78
CRI-tau	19%	26%	2.73

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	20%	40%	1.60
SI	12%	21%	3.39
SSI	7%	20%	3.79
CRI-r	12%	18%	3.63
CRI-rho	12%	18%	3.67
CRI-tau	10%	13%	3.63

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	38%	-30%
PV - SPI	43%	-20%
PV - SCI	43%	-20%
ED - 1	75%	-75%
ED - SPI	43%	-20%
ED - SCI	43%	-20%
ES - 1	9%	-5%
ES - SPI(t)	32%	31%
ES - SCI(t)	32%	31%

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0%	0%
CPI	0%	0%
SPI	5%	5%
SPI(t)	6%	6%
SCI	5%	5%
SCI(t)	6%	6%
0.8 CPI + 0.2 SPI	2%	2%
0.8 CPI + 0.2 SPI(t)	4%	4%

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 11 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	6160.36	-34744.23	-15.4	1.01	0.95	0.88	0.99
std dev	5929.34	40630.85	11.48	0.01	0.05	0.09	0.02
final	16160.00	0.00	-27	1.02	1.00	0.88	1.00

2.3.3.2. Time forecasting

PD	203	Real Duration	220	Late/early	8%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	210.91	9.10	5%	100%
PV - SPI	213.82	13.18	6%	100%
PV - SCI	212.45	13.30	6%	100%
ED - 1	211.09	8.34	5%	100%
ED - SPI	216.45	13.32	5%	100%
ED - SCI	216.18	12.98	5%	100%
ES - 1	218.55	11.45	4%	100%
ES - SPI(t)	232.82	27.39	9%	100%
ES - SCI(t)	232.27	26.89	9%	100%

2.3.3.3. Cost forecasting

BAC	926888.01	Real Cost	910728.010	Over/under budget	1%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	920727.69	5929.33	1%	-1%
CPI	919909.26	6003.65	1%	-1%
SPI	934488.74	23477.03	3%	-3%
SPI(t)	949552.36	39836.27	4%	-4%
SCI	933576.35	22614.91	3%	-3%
SCI(t)	948516.99	38631.44	4%	-4%
0.8 CPI + 0.2 SPI	922528.08	7384.93	1%	-1%
0.8 CPI + 0.2 SPI(t)	924719.85	8647.68	2%	-2%